COMP47350: Data Analytics (Conv)

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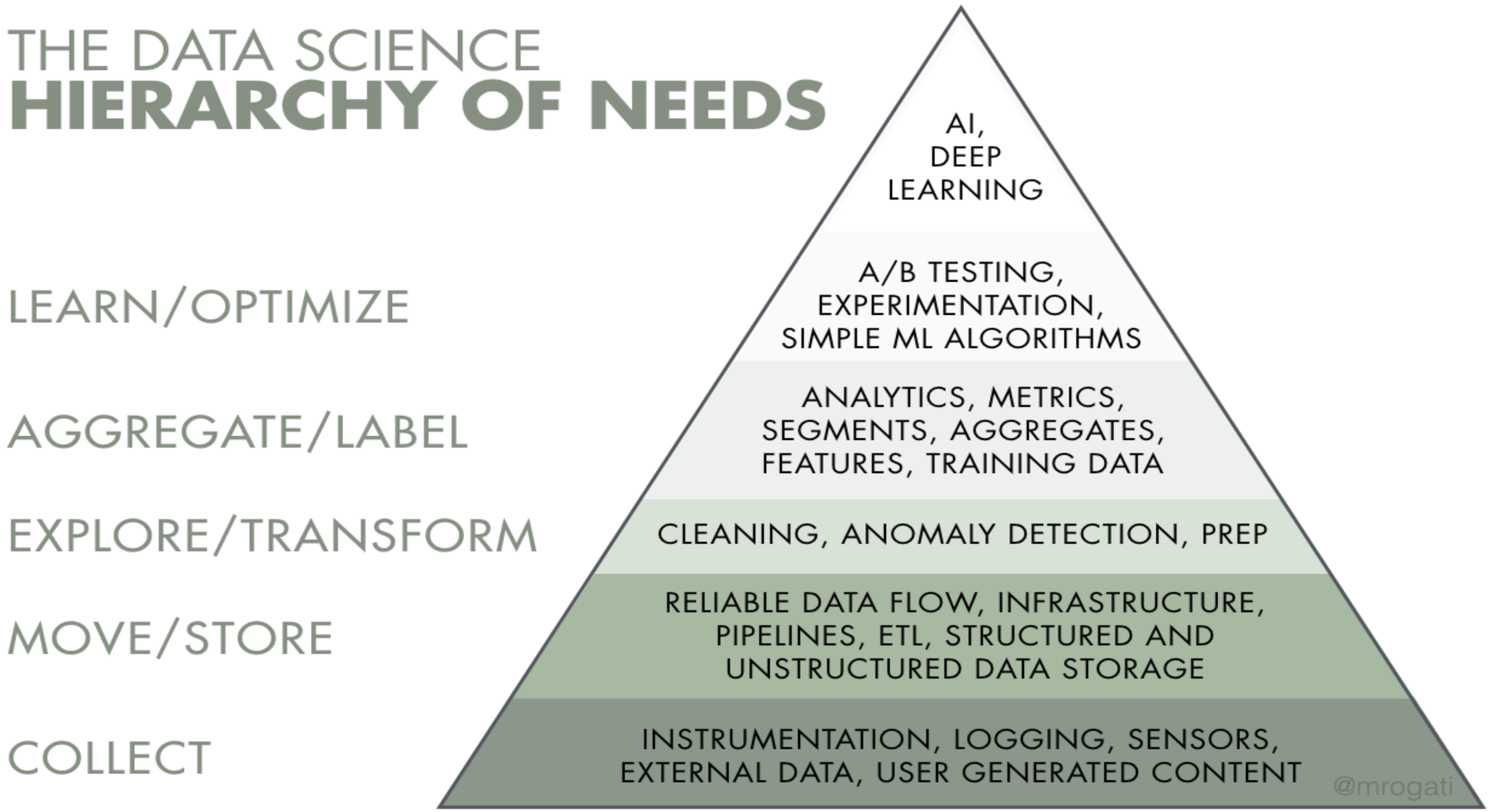
2025/26

Module Topics

- **Python Environment** (Anaconda, Jupyter Notebook)
- **Getting Data** (Web scrapping, APIs, DBs)
- **Understanding Data** (tables, visualisation)
- **Preparing Data** (cleaning, transformation)
- **Modeling & Evaluation** (machine learning)

The DS/AI Hierarchy of Needs



Source: Monica Rogati **"Think of AI as the top of a pyramid of needs. Yes, self-actualization (AI) is great, but you first need food, water and shelter (data literacy, collection and infrastructure)."**

https://hackernoon.com/the-ai-hierarchy-of-needs-18f111fcc007

Data Understanding

- **Data Quality Report (Tables, Visualisations):**
  - Getting To Know The Data
  - Identifying Data Quality Issues
  - Keeping notes about individual features

DQR: A short, written report, supported by tables and visualisations. It is a summary of findings from the data.

Data Quality Report (DQR)

- **Tables: one summary table for continuous** features, **one table for categorical features**
  - We already covered some summary stats for continuous features
  - **This lecture: summary tables** for categorical features and **plots** for both types of features
- **Visualizations:** we plot each feature (barplot, histogram, boxplot)

Data Quality Report

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min.	1st Qrt.	Mean	Median	3rd Qrt.	Max.	Std. Dev.
___	___	___	___	___	___	___	___	___	___	___
___	___	___	___	___	___	___	___	___	___	___
___	___	___	___	___	___	___	___	___	___	___

Cardinality
Missing values
Mean and std deviation
5 stats description of feature values: min, 1st, 2nd, 3rd quartiles, max

Fill the stats for feature Age: [18, 18, 25, 30, 32, 40, 45, 45, 50, 55]

Feature	Min	1st Qrt	Mean	Median	3rd Qrt	Max	Std Dev
Age							

Data Quality Report

(b) Categorical Features

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2^{nd} Mode	2^{nd} Mode Freq.	2^{nd} Mode %
——	——	——	——	——	——	——	——	——	——
——	——	——	——	——	——	——	——	——	——
——	——	——	——	——	——	——	——	——	——

Categorical Features: Descriptive Stats

- **Levels** of a categorical feature = different values that the categorical feature takes. E.g., feature **Country** has levels: Ireland, UK, Germany, etc.

- **Count** of each level = the number of times that level appears in the data sample (level frequency)

- **Proportion** for each level = frequency count for that level divided by the total sample size

- **Note:** We look at both the frequency counts and the proportion (the proportion allows us to compare the feature values across data samples of different size e.g., 100 rows vs 1000 rows)
- Frequencies and proportions are typically presented in a **frequency table** (tabulated data).

Categorical Features: Descriptive Stats

- **Example dataset:** the positions and weekly training expenses of a **school basketball squad** (see textbook)

ID	Position	Training Expenses
1	center	56.75
2	guard	1,800.11
3	guard	1,341.03
4	forward	749.50
5	guard	1,150.00
6	forward	928.30
7	center	250.90
8	guard	806.15
9	guard	1,209.02
10	forward	405.72
11	center	550.00
12	center	223.89
13	center	103.23
14	forward	758.22
15	forward	430.79
16	forward	675.11
17	guard	1,657.20
18	guard	1,405.18
19	guard	760.51
20	forward	985.41

Frequency table for categorical feature **Position**

Level	Count	Proportion
guard	8	40%
forward	7	35%
center	5	25%

Categorical Features: Descriptive Stats

- **Mode =** the most frequent level of a categorical feature
- **Second mode =** the second most frequent level
- For feature **Position** the mode is level **guard,** and the second mode is **forward**

Level	Count	Proportion
guard	8	40%
forward	7	35%
center	5	25%

Motor Insurance Fraud: Dataset ABT

ID	Type	Inc.	Marital Status	Num Clmnts.	Injury Type	Hospital Stay	Claim Amnt.	Total Claimed	Num Claims	Num Soft Tiss.	% Soft Tiss.	Claim Amt Rcvd.	Fraud Flag
1	CI	0		2	Soft Tissue	No	1,625	3250	2	2	1.0	0	1
2	CI	0		2	Back	Yes	15,028	60,112	1		0	15,028	0
3	CI	54,613	Married	1	Broken Limb	No	-99,999	0	0	0	0	572	0
4	CI	0		4	Broken Limb	Yes	5,097	11,661	1	1	1.0	7,864	0
5	CI	0		4	Soft Tissue	No	8869	0	0	0	0	0	1
6	CI	0		1	Broken Limb	Yes	17,480	0	0	0	0	17,480	0
7	CI	52,567	Single	3	Broken Limb	No	3,017	18,102	2	1	0.5	0	1
8	CI	0		2	Back	Yes	7463	0	0	0	0	7,463	0
9	CI	0		1	Soft Tissue	No	2,067	0	0	0	0	2,067	0
10	CI	42,300	Married	4	Back	No	2,260	0	0	0	0	2,260	0
		⋮				⋮					⋮		
300	CI	0		2	Broken Limb	No	2,244	0	0	0	0	2,244	0
301	CI	0		1	Broken Limb	No	1,627	92,283	3	0	0	1,627	0
302	CI	0		3	Serious	Yes	270,200	0	0	0	0	270,200	0
303	CI	0		1	Soft Tissue	No	7,668	92,806	3	0	0	7,668	0
304	CI	46,365	Married	1	Back	No	3,217	0	0		0	1,653	0
		⋮				⋮					⋮		
458	CI	48,176	Married	3	Soft Tissue	Yes	4,653	8,203	1	0	0	4,653	0
459	CI	0		1	Soft Tissue	Yes	881	51,245	3	0	0	0	1
460	CI	0		3	Back	No	8,688	729,792	56	5	0.08	8,688	0
461	CI	47,371	Divorced	1	Broken Limb	Yes	3,194	11,668	1	0	0	3,194	0
462	CI	0		1	Soft Tissue	No	6,821	0	0	0	0	0	1
		⋮				⋮					⋮		
491	CI	40,204	Single	1	Back	No	75,748	11,116	1	0	0	0	1
492	CI	0		1	Broken Limb	No	6,172	6,041	1		0	6,172	0
493	CI	0		1	Soft Tissue	Yes	2,569	20,055	1	0	0	2,569	0
494	CI	31,951	Married	1	Broken Limb	No	5,227	22,095	1	0	0	5,227	0
495	CI	0		2	Back	No	3,813	9,882	3	0	0	0	1
496	CI	0		1	Soft Tissue	No	2,118	0	0	0	0	0	1
497	CI	29,280	Married	4	Broken Limb	Yes	3,199	0	0	0	0	0	1
498	CI	0		1	Broken Limb	Yes	32,469	0	0	0	0	16,763	0
499	CI	46,683	Married	1	Broken Limb	No	179,448	0	0		0	179,448	0
500	CI	0		1	Broken Limb	No	8,259	0	0	0	0	0	1

Data Quality Report: Tables

Motor Insurance Fraud

(a) Continuous Features

Feature	Count	% Miss.	Card.	Min	1st Qrt.	Mean	Median	3rd Qrt.	Max	Std. Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0	11,282.8	729,792.0	35,655.7
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0	1.0	56.0	2.7
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0	0.0	5.0	0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0	0.0	2.0	0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5	8,191.8	295,303.0	30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0	1.0	1.0	0.5

Note: Some errors in the textbook tables vs what we get in the lab when working with Python code and the textbook dataset.

- Example: **FraudFlag** is taken here as a continuous feature, it would make more sense to turn it into categorical.

Data Quality Report: Tables

Motor Insurance Fraud

(a) Categorical Features

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2^{nd} Mode	2^{nd} Mode Freq.	2^{nd} Mode %
INSURANCE TYPE	500	0.0	1	CI	500	1.0	–	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48	24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172	34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146	29.2

Note: More errors in the textbook: Mode% for **InsuranceType** should be 100%.

Note: df.describe() only shows stats for rows without Nan values
df.info() shows how many non-null values each feature has.
Easier to change **Nan** values to **Missing** values and then run df.describe().

Data Quality Report

Continuous features:

- Some issues identified:
  - Income: Median=0 (50% of values are 0)
  - ClaimAmount: Min=-99,999 (negative value for claim)
  - NumSoftTissue: 2% missing data
  - TotalClaim: Max=729k (check row/example with this value)

Categorical features

- Some issues identified:
  - InsuranceType: Card=1 (constant feature)
  - MarritalStatus: 61% missing data (collected together with Income, so Income=0 is actually missing data)

Stats tell us part of the story. To get a more complete view, we need to **plot each feature**.

Data Quality Report

- Tables (continuous features, categorical features)
- **Visualisations** (plots of feature values)

Data Visualisation: Plots

- **Data visualisation** helps to further describe and understand the properties of the data

- Three important techniques to visualize the values of a single feature:
  - **Bar plot (categorical)**
  - **Histogram (numeric)**
  - **Box plot (numeric)**

Data Visualisation

- We will visualise our previous example: basketball players **Position** and **TrainingExpenses**
- We have one categorical and one continuous feature

ID	Position	Training Expenses
1	center	56.75
2	guard	1,800.11
3	guard	1,341.03
4	forward	749.50
5	guard	1,150.00
6	forward	928.30
7	center	250.90
8	guard	806.15
9	guard	1,209.02
10	forward	405.72
11	center	550.00
12	center	223.89
13	center	103.23
14	forward	758.22
15	forward	430.79
16	forward	675.11
17	guard	1,657.20
18	guard	1,405.18
19	guard	760.51
20	forward	985.41

Data Visualisation: Bar plot

- **Bar plot** = counting and plotting the frequency of each value of a feature; **good for plotting categorical features** or continuous (integer) features with cardinality < 10

- A bar per value, the length of each bar visually represents the frequency count

- Feature **Position**

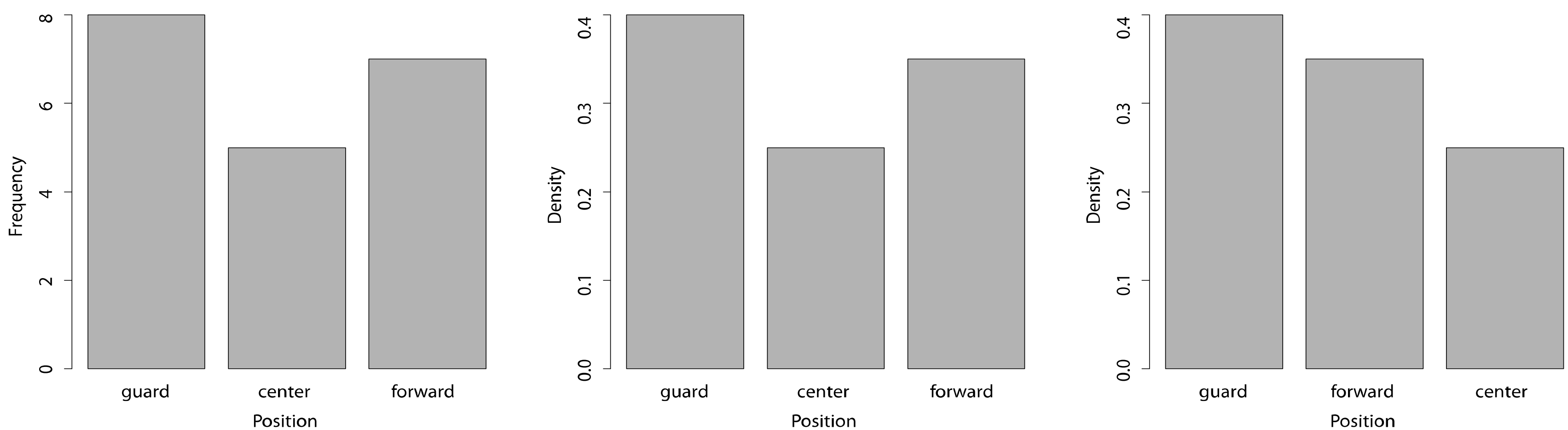
Level	Count	Proportion
guard	8	40%
forward	7	35%
center	5	25%

Probability of value 'guard' in this sample

Data Visualisation: Bar plot

Bar plot for feature **Position** (left plot uses frequency counts; middle plot uses proportions, i.e., probabilities)

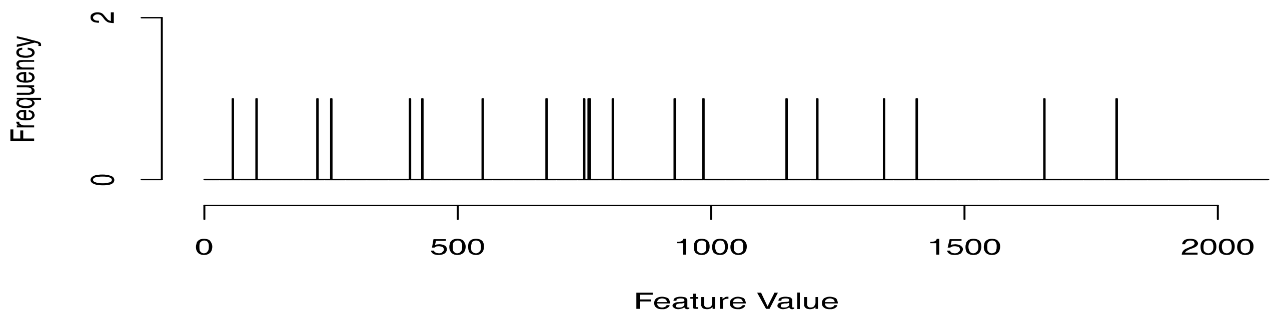
Bar plots are great for categorical features



(a) Frequency (b) Proportion (c) Ordered

Data Visualisation: Bar plot

Bar plots don't work for continuous features



For continuous features: we need to first group the values into intervals, then count the values falling in each interval (bin)

Example: values 4.5, 4.6, 5.0 fall in **interval [0,10),**
values 10.5, 11.6, 13.9 fall **in interval [10,20), …,**

We sort the continuous numeric values and break them up into intervals, aka bins. The plot of these counts is called a **histogram.**

Data Visualisation: Histograms

- For continuous features we use plots called **histograms**

ID	Position	Training Expenses
1	center	56.75
2	guard	1,800.11
3	guard	1,341.03
4	forward	749.50
5	guard	1,150.00
6	forward	928.30
7	center	250.90
8	guard	806.15
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10	forward	405.72
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16	forward	675.11
17	guard	1,657.20
18	guard	1,405.18
19	guard	760.51
20	forward	985.41

Data Visualisation: Histograms

- First divide the range of a variable into **equal-length intervals**, (aka bins).
- Split the values of feature **TrainingExpense** (min = 56.75, max = 1,800.11) into 10 equal-length intervals.
- Need to decide how many bins we want to use (typically 10 bins are used by default)

- This approach at plotting the distribution of values for a feature using equal-length intervals and counting the values which fall in each interval, is called an **equi-width binning histogram.**

- There are other ways to computes histograms but they are less intuitive and hence less common (eg equi-depth histogram used in databases: break data into intervals with the same number of values).

Histogram Example

20 rows

ID	Position	Training Expenses
1	center	56.75
2	guard	1,800.11
3	guard	1,341.03
4	forward	749.50
5	guard	1,150.00
6	forward	928.30
7	center	250.90
8	guard	806.15
9	guard	1,209.02
10	forward	405.72
11	center	550.00
12	center	223.89
13	center	103.23
14	forward	758.22
15	forward	430.79
16	forward	675.11
17	guard	1,657.20
18	guard	1,405.18
19	guard	760.51
20	forward	985.41

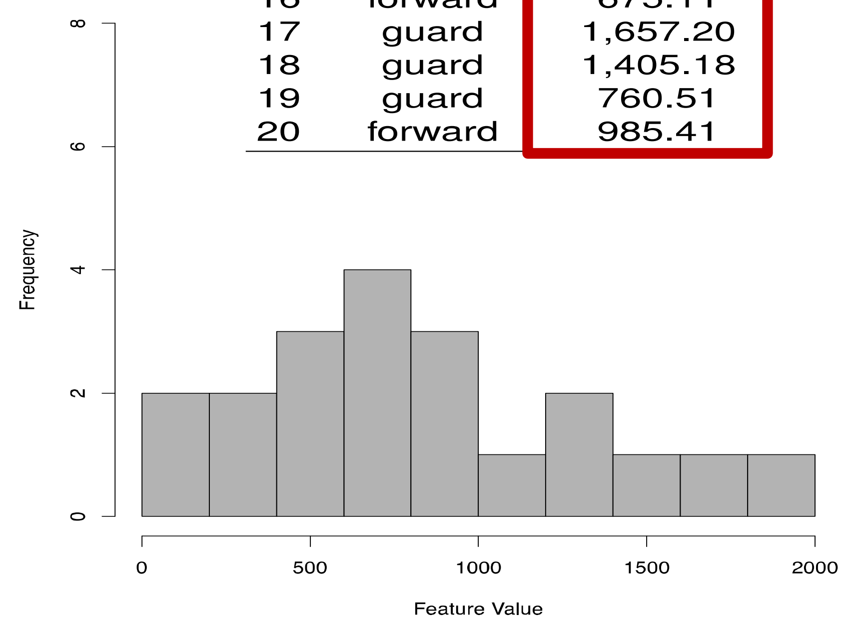
Histogram of **TrainingExpenses** with 10 bins

For bin [0, 200):

- Prob[0,200) = 2/**20** = 0.1
- Density = prob /width
- Density for [0,200) = 0.1/200 = 0.0005
- Density allows to compare histograms that use different bin widths

(a) 200 unit intervals

Interval	Count	Density	Prob
$[0, 200)$	2	0.0005	0.1
$[200, 400)$	2	0.0005	0.1
$[400, 600)$	3	0.00075	0.15
$[600, 800)$	4	0.001	0.2
$[800, 1000)$	3	0.00075	0.15
$[1000, 1200)$	1	0.00025	0.05
$[1200, 1400)$	2	0.0005	0.1
$[1400, 1600)$	1	0.00025	0.05
$[1600, 1800)$	1	0.00025	0.05
$[1800, 2000)$	1	0.00025	0.02

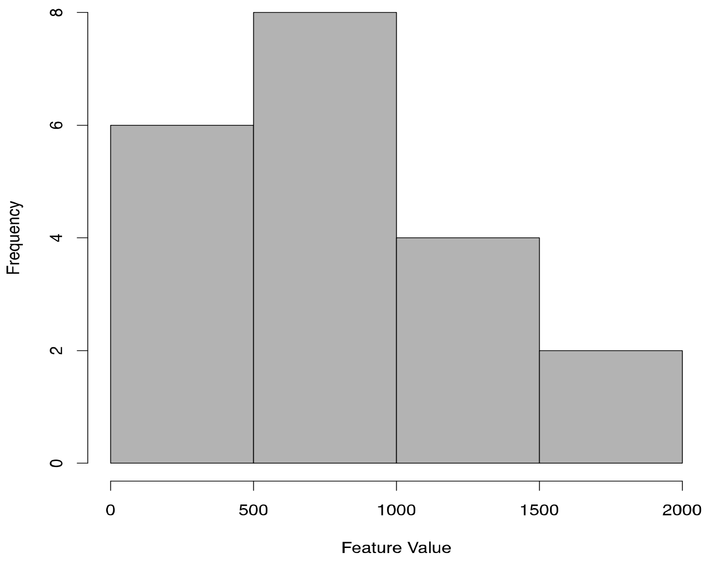


Data Visualisation: Histograms

- Changing the number of bins: zoom in/out of the data
- Histogram of **TrainingExpenses** with 4 bins (intervals broken at 500 units)
- The bar width shows interval size, the bar height shows how many values fall in the interval

(b) 500 unit intervals

Interval	Count	Density	Prob
[0, 500)	6	0.0006	0.3
[500, 1000)	8	0.0008	0.4
[1000, 1500)	4	0.0004	0.2
[1500, 2000)	2	0.0002	0.1



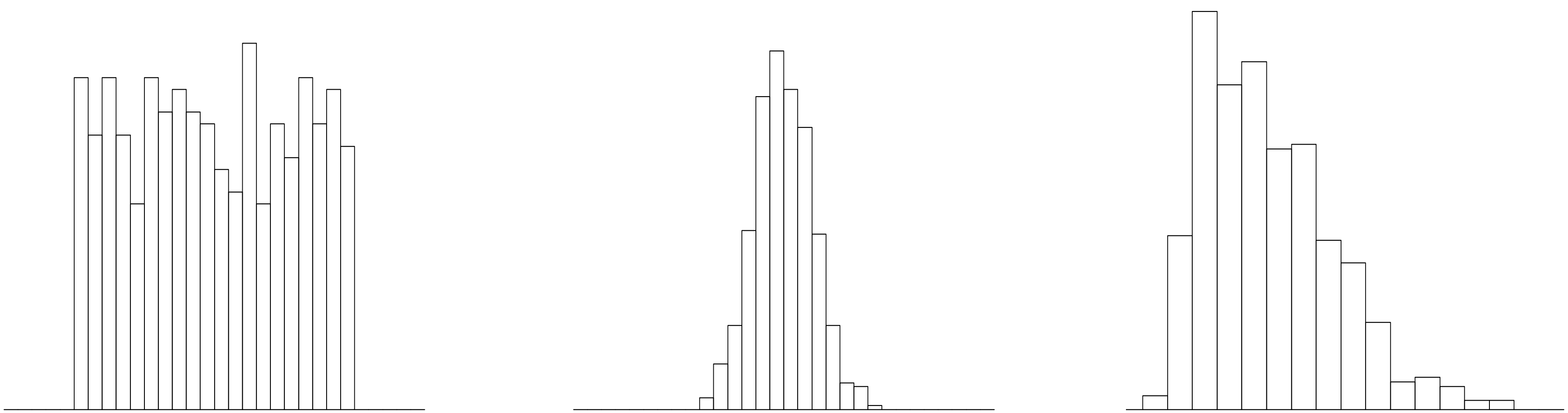
Data Visualisation: Histograms

- Histograms give us information on the probability distribution of a sample of values

- When we generate **histograms** of features, there are some **common, well understood shapes** that we should look out for

Histograms & Probability Distributions

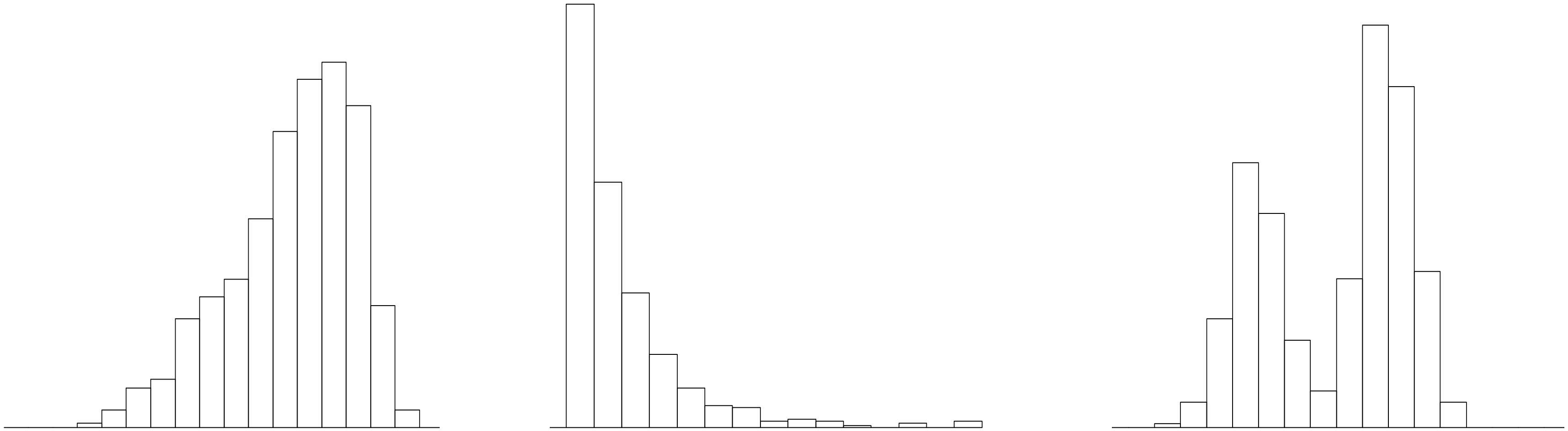
- Common distribution shapes



(a) Uniform (b) Normal (Unimodal) (c) Unimodal (skewed right)

Histograms & Probability Distributions

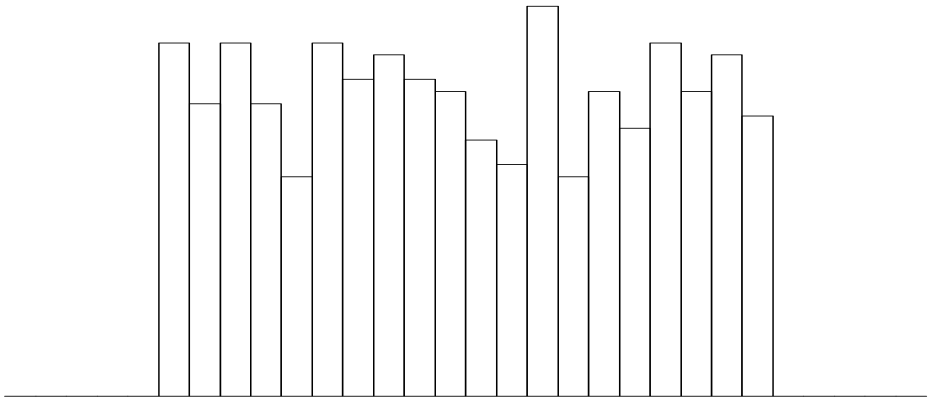
- Common distribution shapes



(a) Unimodal (skewed left)   (b) Exponential   (c) Multimodal

Histograms & Probability Distributions

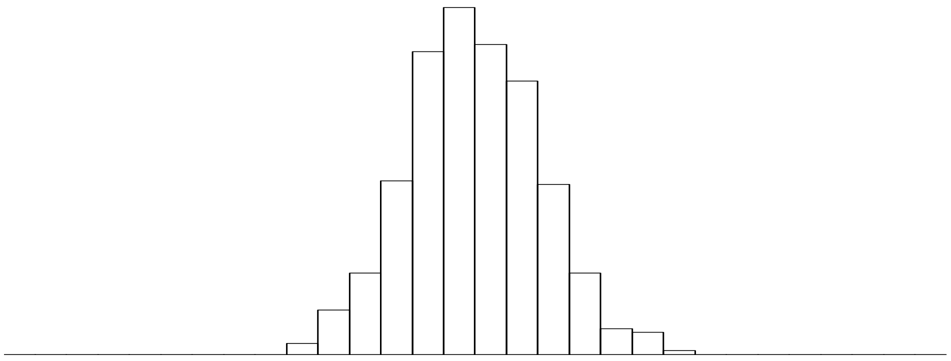
- **Uniform distribution:** indicates that a feature is equally likely to take a value in any of the ranges present (e.g., the feature ID); **usually not a very informative feature from the point of view of target prediction**



Uniform

Histograms & Probability Distributions

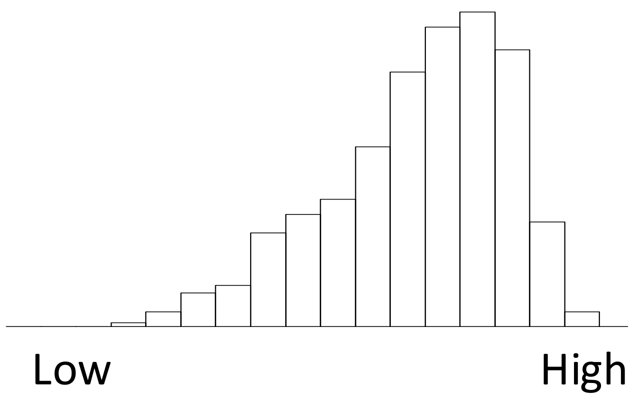
- **Normal (Gaussian) distribution:**
  - Features have a strong tendency towards a central value and symmetrical variation to either side of this value
  - E.g., heights of randomly selected groups of women or of men; men height varies around the mean of 170cm; women height varies around the mean of 160cm; height distribution is Gaussian (aka bell shaped)
  - Lots of natural phenomena are Gaussian distributed (Central Limit Theorem)



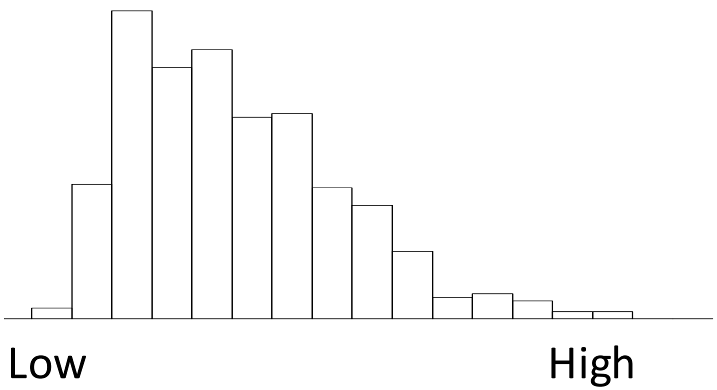
Normal (Unimodal)

Histograms & Probability Distributions

- **Unimodal (skewed) distribution:** Skew is a tendency towards very low or very high values (e.g., skewed left: age of death from natural causes; skewed right: salary, a few people have very high salaries leading to a long tail to the right)



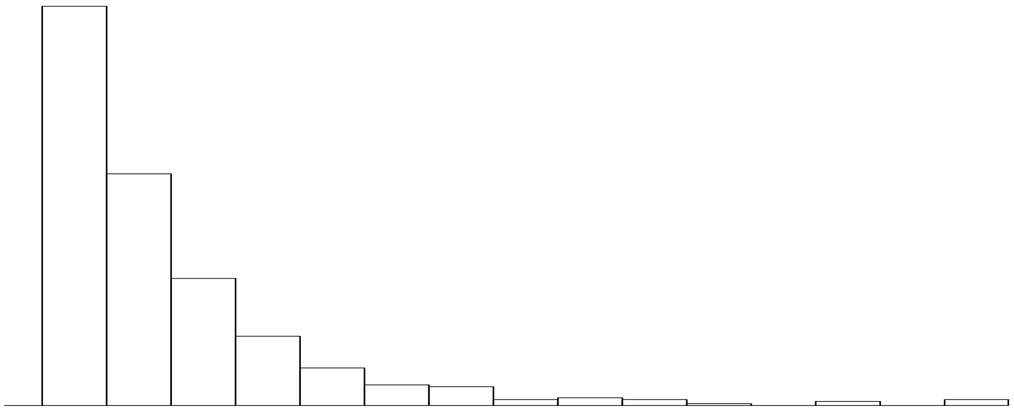
Unimodal (skewed left)



Unimodal (skewed right)

Histograms & Probability Distributions

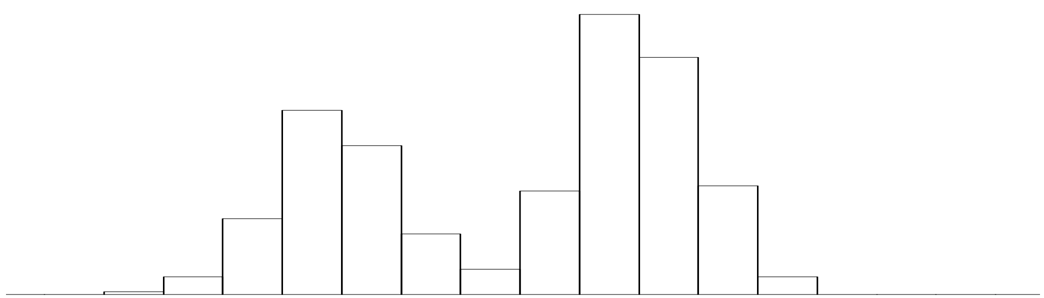
- **Exponential distribution:** the likelihood of occurrence of a small number of low values is very high, but sharply diminishes as values increase (e.g., number of times a person has been married, it is common for 0, 1, 2 times very uncommon for 8 times or more)



Exponential

Histograms & Probability Distributions

- **Multimodal distribution:** A feature with a **multimodal distribution** has two or more commonly occurring ranges of values that are clearly separated (e.g., the heights of women and men taken together will have two separate modes, looks like 2 separate Gaussians)
- Central tendency and variation stats tend to break down
- Multimodal distribution may help separate different groups, so it tends to be useful for prediction



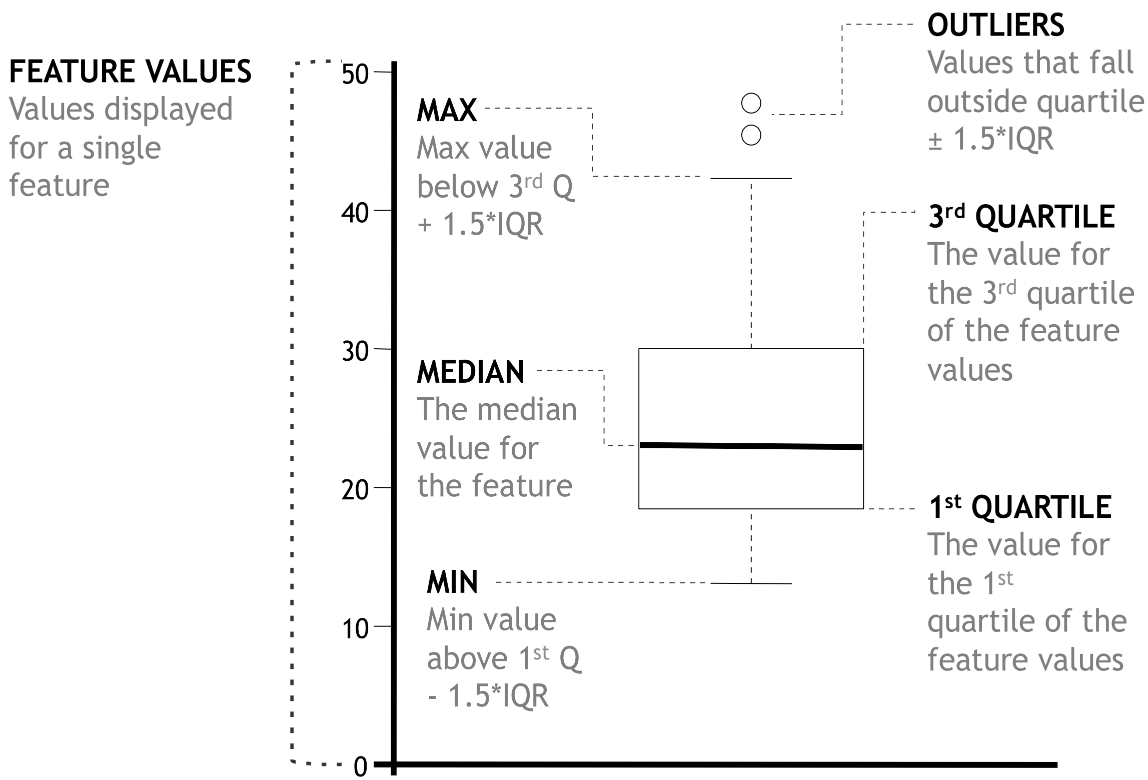
Multimodal

Populations & Samples

- It is very important to understand the difference between a **population** and a **sample**
- **Population** = all possible measurements or outcomes that are of interest to us in a particular analysis (e.g., all people in Ireland with voting rights)
- **Sample** = the subset of the population that is selected for analysis (e.g., people in this class with voting rights). We typically work with samples to try to understand properties of a population
- **The sample needs to be representative of the population and large enough to allow accurate estimates for descriptive stats and prediction**

Data Visualisation: Box plot

- **Box plots: visual representation of spread of feature values** and **5 popular stats** for continuous features (min, 1st quartile, median, 3rd quartile, max)



Outliers: use with caution, this is a blunt tool. The definition of an outlier usually depends on the problem domain.

Useful to also check other percentiles.

Group Project

- Project starts this week (details on Brightspace).

- If you haven't submitted your group project members, you cannot get your dataset. Deadline to submit group members was Feb 6, 2026. Submit today on Brightspace.

- Make sure that your code runs and that all results can be reproduced based on your data **(note: go through checklist on Brightspace before submission).**

- Questions?
- At lab, lecture and on Brightspace/Discussion.

References

- **FMLPDA Book: Fundamentals of Machine Learning for Predictive Data Analytics**, John D. Kelleher, Brian Mac Namee, Aoife D'Arcy, MIT Press, 2015

  Appendix A: Descriptive Stats and Visualisation (machinelearningbook.com)